Oscillator: It generates high-frequency carrier wave (a sinusoidal wave) for long distance transmission into space. As the energy of a wave is proportional to its frequency, the carrier wave has very high energy.

Modulator: It superimposes the baseband signal onto the carrier signal and generates the modulated signal.

Power amplifier: It increases the power level of the electrical signal in order to cover a large distance.

iv) Transmitting antenna

It radiates the radio signal into space in all directions. It travels in the form of electromagnetic waves with the speed of light.

v) Communication channel

Communication channel is used to carry the electrical signal from transmitter to receiver with less noise or distortion. The examples for communication medium are wires, cables, optical fibres in wireline communication and free space in wireless communication.

vi) Receiver

The signals that are transmitted through the communication medium are received by a receiving antenna which converts EM waves into RF signals and are fed into the receiver.

The receiver consists of electronic circuits like demodulator, amplifier etc. The demodulator extracts the baseband signal from the modulated signal. Then the baseband signal is amplified using amplifier. Finally, it is fed to the output transducer.

vii) Repeaters

Repeaters are used to increase the range or distance through which the signals are sent. It is a combination of transmitter and receiver. The signals are received, amplified and retransmitted with a carrier signal of different frequency to the destination. The best example is the communication satellite in space.

viii) Output transducer

It converts the electrical signal back to its original form such as sound, music, pictures or data. Examples of output transducers are loudspeakers, picture tubes, computer monitor etc.

10.10.1 BASIC DEFINITIONS IN ELECTRONIC COMMUNICATION SYSTEM

For better understanding of the communication systems, it is essential to know about the following terms.

i) Range

It is the maximum distance between the source and the destination up to which the signal is received with sufficient strength.

ii) Noise

It is the undesirable electrical signal that interferes with the transmitted signal. Noise attenuates or reduces the quality of the transmitted signal. It may be man-made (automobiles, welding machines, electric motors etc.) or natural (lightning, environmental effects, radiation from sun and stars). Noise cannot be completely eliminated. However, it can be reduced using various techniques.

iii) Attenuation

The loss of strength of a signal while propagating through a medium is known as attenuation.

iv) Bandwidth

The frequency range over which the baseband signals or the information

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